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1: Which approach to AI focuses on designing systems that behave in a way indistinguishable from human behavior?

- A) Acting humanly
- B) Thinking humanly
- C) Thinking rationally
- D) Acting rationally

2: What best defines a rational agent in AI?

- A) An agent that always chooses actions that maximize its expected performance measure based on available information.
- B) An agent that mimics human decision-making, even if it is not optimal.
- C) An agent that learns from past experiences but does not necessarily act optimally.
- D) An agent that operates without any predefined goals or objectives.

3: In the context of AI, what is the primary role of sensors and actuators in an agent?

- A) Sensors store internal knowledge, while actuators process logical reasoning.
- B) Sensors gather information from the environment, while actuators execute actions to affect the environment.
- C) Sensors are used for decision-making, while actuators analyze sensory inputs.
- D) Sensors communicate with other agents, while actuators perform computations.

4: What is a percept sequence in the context of an intelligent agent?

- A) A sequence of actions taken by an agent to achieve its goal.
- B) A history of all percepts received by an agent up to a given point in time.
- C) A set of rules that determine an agent's behavior in an environment.
- D) A sequence of commands sent by an agent to its actuators.

5: Which of the following best describes the difference between a single-agent and a multi-agent environment in AI?

- A) A single-agent environment consists of multiple agents that do not interact, whereas a multi-agent environment has interacting agents.
- B) A single-agent environment involves only one decision-making entity, whereas a multi-agent environment includes multiple entities that may compete or cooperate.
- C) A single-agent environment always has full observability, while a multi-agent environment is always partially observable.
- D) A single-agent environment follows a deterministic model, whereas a multi-agent environment follows a stochastic model.

6: What is the key distinction between an episodic and a sequential environment in AI?

- A) In an episodic environment, an agent's actions affect future decisions, while in a sequential environment, each decision is independent.
- B) In an episodic environment, the agent makes decisions based on a fixed set of rules, whereas in a sequential environment, the agent adapts to changing conditions.
- C) In an episodic environment, each action is independent of actions of previous episodes, whereas in a sequential environment, current actions influence future actions.
- D) In an episodic environment, the agent interacts with multiple agents, while in a sequential environment, it operates alone.

7: In the context of search problems in AI, what is the primary role of the transition model?

- A) It determines when a state satisfies the goal conditions.
- B) It specifies the cost associated with moving from one state to another.
- C) It defines the result of applying an action to a given state, producing the next state.
- D) It lists all possible actions available in a given state.

8: Which of the following best characterizes uninformed search strategies in AI?

- A) They use problem-specific knowledge to guide the search toward the goal more efficiently.
 - B) They explore the search space without using any domain-specific information.
 - C) They guarantee finding the optimal solution in the minimum possible time.
 - D) They require a heuristic function to evaluate the quality of each state.
- 9: Which of the following is a key characteristic of Breadth-First Search (BFS)?**
- A) BFS explores the deepest nodes in the search tree before expanding shallower nodes.
 - B) BFS finds the shortest path in terms of the number of actions.
 - C) BFS is a heuristic-based search algorithm that prioritizes promising paths.
 - D) BFS is an uninformed search that does not guarantee completeness.
- 10: Which of the following best describes Depth-First Search (DFS)?**
- A) DFS explores all the nodes at the current depth before moving on to the next level.
 - B) DFS explores the deepest nodes first, backtracking when it reaches a dead end.
 - C) DFS guarantees finding the shortest path to the goal, if one exists.
 - D) DFS is guaranteed to find a solution even in infinite state spaces.
- 11: What is the main advantage of Uniform Cost Search (UCS) over other search algorithms?**
- A) UCS guarantees finding the solution with the least number of actions, regardless of the action costs.
 - B) UCS always finds the optimal solution by expanding the least costly path first.
 - C) UCS is faster than other search algorithms because it uses a heuristic to guide the search.
 - D) UCS is a depth-first algorithm that explores deeper nodes first.
- 12: What distinguishes an informed search algorithm from an uninformed search algorithm in AI?**
- A) Informed search algorithms use domain-specific knowledge to guide the search process.
 - B) Informed search algorithms explore all possible paths regardless of their costs or distances.
 - C) Informed search algorithms are guaranteed to find the shortest path in the least amount of time.
 - D) Informed search algorithms do not use any information about the goal or the state space.
- 13: What is the primary role of a heuristic function in search algorithms?**
- A) It defines the exact solution to the problem, eliminating the need for search.
 - B) It estimates the cost or distance from a given state to the goal state, guiding the search.
 - C) It evaluates all possible actions and their outcomes to guarantee the best solution.
 - D) It randomly selects actions from the available options to explore the state space.
- 14: Why does A* search algorithm not require a visited set to avoid revisiting nodes?**
- A) A* considers the cost from the start node to the current node.
 - B) A* uses a heuristic that automatically prevents revisiting any state.
 - C) A* never requires backtracking.
 - D) A* explores all nodes optimally without checking for previously visited nodes, relying on the heuristic.
- 15: What is a key characteristic of adversarial search in AI?**
- A) It involves searching for an optimal solution in a single-agent environment without considering opponents.
 - B) It is used to model environments where agents or players make decisions to maximize their own payoff while minimizing the opponent's payoff.
 - C) It focuses on finding solutions with the least possible cost without considering the adversary's actions.
 - D) It is an uninformed search method that does not use any heuristics or domain-specific knowledge.
- 16: What is the primary advantage of using alpha-beta pruning in adversarial search?**
- A) It guarantees finding the optimal move, but it explores all possible moves.
 - B) It eliminates the need to explore branches of the search tree that cannot affect the final decision.
 - C) It expands the entire search tree, ensuring that all possible outcomes are considered.
 - D) It only works with deterministic environments where actions have no uncertainty.
- 17: What is the main feature of local search algorithms in AI?**
- A) Local search algorithms explore the entire search space exhaustively, evaluating all possible states.
 - B) Local search algorithms operate by evaluating the entire problem space, ensuring an optimal solution

is found.

- C) Local search algorithms focus on finding a solution by iteratively improving a single state, without considering the full search tree.
- D) Local search algorithms are guaranteed to find the optimal solution in all problem domains.

18: What is the key idea behind the Simulated Annealing algorithm?

- A) It explores the search space by randomly selecting neighboring solutions and accepting worse solutions with a decreasing probability over time.
- B) It guarantees finding the global optimum by exhaustively searching the entire solution space.
- C) It only accepts better solutions and moves toward the best possible solution in a greedy manner.
- D) It works by evaluating all possible solutions and selecting the one with the lowest cost.

19: What is the role of crossover in a genetic algorithm?

- A) Crossover combines two parent solutions to create offspring by exchanging parts of their genetic material.
- B) Crossover selects the best solution from the population and discards the rest.
- C) Crossover mutates the solutions to introduce randomness and avoid premature convergence.
- D) Crossover evaluates the fitness of solutions and uses this to guide the search.

20: What is a key characteristic of a Constraint Satisfaction Problem (CSP) in AI?

- A) CSPs involve finding a solution without backtracking.
- B) CSPs focus on optimizing a performance measure by assigning values to variables.
- C) CSPs require finding values for variables that meet all specified constraints.
- D) CSPs are always solvable through local search techniques without considering constraints.

21: What does consistency refer to in the context of CSP?

- A) Consistency ensures that the solution to the problem is optimal according to a given performance measure.
- B) Consistency means that the values assigned to variables do not violate any constraints, such as unary or binary constraints.
- C) Consistency allows the problem to be solved using any heuristic without considering the constraints.
- D) Consistency refers to the ability of the CSP to be solved by exhaustive search techniques.

22: What is the primary goal of the AC-3 algorithm in CSP?

- A) AC-3 ensures that the solution to the CSP is found by exploring all possible variable assignments.
- B) AC-3 makes the CSP arc-consistent by iteratively removing inconsistent values from the domains of variables.
- C) AC-3 guarantees an optimal solution by using a backtracking search with heuristics.
- D) AC-3 focuses on minimizing the number of constraints in the problem.

23: What is the purpose of the Minimum Remaining Values (MRV) heuristic in CSP?

- A) MRV selects variables that have the fewest constraints, aiming to reduce the solution space.
- B) MRV chooses variables with the smallest domain size, prioritizing those with the fewest remaining possible values.
- C) MRV selects variables that are least constrained by the current assignments, ensuring faster problem-solving.
- D) MRV is used to determine the optimal order in which to assign values to the variables.

24: What is the primary objective of the Least Constraining Value (LCV) heuristic in CSP?

- A) LCV chooses the value that leaves the largest number of valid options for other variables.
- B) LCV selects the value that satisfies the most constraints in the problem.
- C) LCV chooses the value that is most likely to violate future constraints, ensuring a thorough search.
- D) LCV prioritizes the value that reduces the number of possible solutions for the remaining variables.